

operating in chunks of 128 bits instead. By changing the perspective, we would actually be simplifying the algorithm. But then again, the simplification would only be in terms of thinking, not in terms of writing it as a program for the computer.

A view from the implementation perspective

We said earlier that implementations of any algorithm on a machine are limited by the capabilities of the machine and that 32-bit machines can take in only 32 bits and 64-bit machines can take only up to 64-bits in one cycle. Hence, when implementing this algorithm on the machine, you cannot just stuff up all the 128-bits of the plaintext in a variable and 128-bits of the key into another variable and run the XOR operation. The XOR operation is always done on bits and the number of bits that a machine can take in is limited. Hence, your implementation as such will have to consider those boundaries. In such a context, you will have to go with either chunks of 32 bits or 64 bits depending on the machine on which you're going to run the encryption.

It's the same reason why we have different setup files for 32-bit and 64-bit architectures for the same version of the same software. The original perspective of running the XOR operation in blocks of 32-bit was not all that bad. After all that's how you would be doing it on a 32-bit machine.

Suitable programming languages for implementation

Just because you can create beautiful web pages with great functionality using PHP and Javascript does not mean you can use those for implementing the cipher we just designed. Some factors which would govern the decision on the programming language to be used for implementation are:

1. **Features provided by the language:** While PHP and Javascript can be used to do bit-level operations, those are not the things these languages were designed for. The use of PHP and Javascript remains on a higher level than playing with bits. A little lower level programming language such as Java, C, C++, etc., would be what you need.
2. **Speed:** PHP and Javascript are not really programming languages in the classical sense. These are scripting languages. Though you do write programs in them and they do work, the programs are not compiled. They are interpreted. While Java has a rich range of functionality and can work on bit-levels, it is not really fast at doing that (after all, Java programs